



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	POLYMER CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basic understanding of organic chemical reactions and molecular structure.				
Course Summary	The course covers polymers comprehensively, including classification, polymerization methods, properties, processing, and commercial applications. Students classify polymers by origin, synthesis, and structure, analyzing chain and step growth polymerizations. They apply polymer property knowledge to control performance and explore processing techniques like bulk and suspension polymerizations, calendering, and injection molding. Additionally, students evaluate commercial polymers for industrial use, gaining a thorough understanding of their importance in various industries.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the classification of polymers based on origin, synthesis, structure, and intermolecular forces.	U	F	Instructor-created exams / Quiz /Assignment
CO2	Analyze chain and step growth polymerizations, discerning mechanisms and factors influencing polymerization processes.	An	C	Class test /Assignment /Quiz
CO3	Apply knowledge of polymer properties such as molecular weights, viscosity, and rheological behaviour to predict and control polymer performance.	Ap	C	Assignment/ Class test

CO4	Understand various polymer processing techniques including bulk, solution, and suspension polymerizations.	U	C	Assignments /Seminar presentation
CO5	Evaluate the suitability of different commercial polymers for specific industrial applications, utilizing understanding of polymer properties and processing techniques	E	C	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Polymers Types of Polymerisation		12	20
	1	Polymers and macromolecules – Monomers – Homo and hetero polymers – Copolymers.	1	
	2	Classification based on origin (natural, semi synthetic and synthetic) and synthesis (addition and condensation).	2	
	3	Classification based on structure (linear, branched chain and cross linked) and intermolecular forces (elastomeres, fibres, thermoplastics and thermosetting polymers)	2	
	4	Tacticity in polymers- polymer chain flexibility- factors affecting chain flexibility	2	
	5	Glass transition temperature and crystalline melting points- variation and structures- molecular interpretation of glassy state of polymers	2	
	6	Chain and step growth polymerizations – Free radical, ionic and coordination polymerizations with mechanism – Zeigler-Natta polymerization	3	
II	Properties of Polymers		12	20
	7	Molecular weights of polymers: Average molecular weights – Number average and Weight average molecular weights	2	
	8	Sedimentation average (Method of determination not required) and Viscosity average molecular weight – determination of viscosity average molecular weight	3	
	9	Polydispersity index and molecular weight distribution; Molecular weight and Degree of polymerization.	3	
	10	Introduction to polymer melt rheology Newtonian fluids- non-Newtonian fluids. Bingham plastics, pseudo plastics- rheopectic and thixotropic behaviour- rheological measurements	4	

III	Polymerisation Techniques and Polymer Processing		12	20
	11	Polymerisation Techniques: Bulk, solution, suspension, emulsion, melt condensation and interfacial polycondensation polymerizations.	2	
	12	Calendering, rotational moulding, compression, injection moulding, blow moulding and thermoforming.	1	
	13	Additives for compounding rubbers- mastication, two roll milling, internal mixing, compounding ingredients, pigments.	2	
	14	Processing aids- processing methods for manufacture of products: blending, calendaring, extrusion and moulding.	3	
	15	Different elastomer curing systems: efficient, semi efficient, conventional and sulphurless cure mechanism of vulcanization, sulphur vulcanizing systems, non-sulphur vulcanizing systems for olefin rubbers	3	
	16	Polymer composites, Properties and its different types- Process of tailoring properties	1	
IV	Polymer Testing and Commercial Polymers (12 hrs)		12	20
	17	Importance of standards and standard organizations- processability and performance- testing of plastics and rubbers material characterization tests such as hardness, tensile stress/strain, compression stress/strain, shear stress/strain, flexural stress/strain, tear tests, rebound resilience, friction, creep, fatigue.	2	
	18	Pollution due to plastics – Recycling of plastics - Plastic identification codes.	2	
	19	Preparation, Structure, properties and applications of: Polyolefins (HDPE, LDPE, PP and PS); Vinyl polymers (PVC, PVP and EVA, Saran); fluoro polymers (Teflon); Acrylic polymers (PAN and PMMA)	2	
	20	Preparation, Structure, properties and applications of: Aromatic polyamides: (kevlar); Polyester (terylene); Polycarbonate (lexan); Polyurethanes; Resins- Glyptal and formaldehyde resins (UF, MF and PF).	2	
	21	Preparation, Structure, properties and applications of: Rubbers (natural rubber, silicone rubber, and polyurethane elastomers, EPDM, BR, SBR, nitrile rubber, Neoprene, Butyl rubber).	2	
	22	Preparation, Structure, properties and applications of: Fibers: (nylon 66 and nylon 6,). Adhesives: (cyanoacrylate, epoxy adhesives, and polyvinyl acetate (PVA) adhesives). Biodegradable Polymers (polylactic acid (PLA) and polyhydroxyalkanoates (PHA)). Conductive Polymers (polyaniline and polyacetylene- concept of doping.	2	
V	Experiments in Polymer Chemistry (12 hrs)		12	20
	Open Ended	Synthesis of Polyaniline Synthesis of Phenol Formaldehyde Resin	12	

		Molecular weight determination using viscometric method Problem solving related to molecular weight calculation		
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References:

1. F. W. Billmeyer Jr., Textbook of Polymer Science, John Wiley and Sons, New Delhi, 2007.
2. V. R. Gowarikar, Polymer Chemistry, New Age International Pvt. Ltd., New Delhi, 2010.
3. B. K. Sharma, Polymer Chemistry, Goel Publishing House, Meerut, 1989.
4. M. G. Arora, M. Singh, M. S. Yadav, Polymer Chemistry, 2nd Revised Edn., Anmolpublications Private Ltd., New Delhi, 1989.
5. K. J. Saunders, Organic Polymer Chemistry, 2nd Edn., Chapman and Hall, London, 1988.
6. Malcolm P. Stevens, Polymer Chemistry: An Introduction, 3rd Edn., Oxford University Press, USA, 1998.
7. Gowri Sankar Misra, Introductory Polymer Chemistry, New Age International, New Delhi, 1993.
8. M. S. Bhatnagar, Polymer Chemistry, S Chand and Company Pvt. Ltd., New Delhi, 2014 Further reading
9. R. B. Seymour, C. E. Carraher, Polymer Chemistry: An Introduction, Marcel Dekker, Inc. New York, 1981.
10. G. Odian, Principles of Polymerization, 4th Edn., Wiley, 2004.
11. P. Ghosh, Polymer Science & Technology, Tata McGraw-Hill Education, 1991.
12. R. W. Lenz, Organic Chemistry of Synthetic High Polymers, Interscience Publishers, New York, 1967.
13. M. P. Stevens, Polymer Chemistry: An Introduction, 3rd Edn., Oxford University Press, 2005.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2						3		2		1		
CO 2	2				1	2	3		3		1		1
CO 3				1	3	1	3		3		1		
CO 4				2	3	1	3		3		1		1

CO 5				3		3	3		2		1		1
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓